

1 Schedule Controlling

Overview

Menu	Order management → Order controlling → Schedule controlling
Transaction code	scec
Function authorization	scec

Available user fields		
Where?	Object type/user field key	Source (type)
Table Schedule controlling	AUNR/SYSTEM	Order (MF-D)
Table Schedule controlling	AGNR/SYSTEM	Operation (MF-D)
How to configure user fields? Which user field types are available?		

Purpose

You use this application to monitor deviations from scheduled dates for operations. The deviations are displayed in a list. The list provides answers to the following questions, for example:

- Which operations whose planned start has been exceeded have not (yet) been started?
- Which operations are currently not being processed (status prepared or interrupted) and are expected to be delayed as a result?
- Which operations that are currently being processed (status running), are expected to be finished too late?
- Which operations were finished too late?

You can use this function to analyze lateness in the schedule, but you can also see which order start or order end is too early.

Target group for this type of information: production controllers and supervisors. Benefit of the function: The quality of the scheduled times is continuously monitored. It is a tool to ensure smooth production processes.

Integration

During the analysis, current dates are compared to planned dates. Depending on the type of planning, the function can monitor dates planned in detail, scheduled dates or basic dates. The planned dates are either the result of detailed planning performed in HYDRA shop floor scheduling or of the planning performed in the ERP system.

Requirements

To use this application properly, you must know how the planned dates are identified or set in your system. Only then you can specify a useful selection.

Selection criteria

The application provides the following selection criteria:

Check against baseline plan

The selection list shows all baseline plans that are included in the responsibility area the user is authorized for.

OPs started too early, OPs started too late, OPs finished too early, OPs finished too late

Depending on the selection options checked, the following operations are included in the selection:

- Operations started too early
- Operations finished too early
- Operations started too late
- Operations finished too late

The values that are calculated are specified by the- option selected: "Exceeding of planned dates", "Exceeding of basic dates" or "Exceeding of scheduled dates".

If none of these options is set, all operations without deviations are displayed.

Date ... to ...

The system only checks the operations that have a start or end date in the specified period (depending on the selected option).

Consider long-term data

If this option is set, the system also includes operations that have been transferred from the online data to the long-term dataset.

Exceeding of planned dates

The system checks if the start and end dates specified in the detailed scheduling (e.g. HYDRA Shop Floor Scheduling) are respected.

Exceeding of basic dates

The system checks if the basic dates *Latest start* (LST) and *Latest end* (LET) transferred from the ERP system or specified in the HYDRA lead time scheduling are respected.

Note:

The ERP system must transfer the correct basic dates for the operation that result from lead time scheduling. Other option: You can also calculate the basic dates using the HYDRA lead time scheduling.

Exceeding of scheduled dates

The system checks if the calculated dates *Scheduled start time* and *Scheduled end time* transferred from the ERP system or specified in the HYDRA lead time scheduling are respected.

Planned for

Workplace or group

Workplace/group/cost center/company

Narrows down the display by workplace, group or cost center. You can also use wildcards.

Note: The selection by workplace or cost center is only useful with operations that are already planned for a specific workplace.

Order

Only operations of a specific order are selected. You can also use wildcards.

Order type

Only operations of orders of a specific order type are selected. Multiple selection is possible.

Category

Only operations of orders of a specific order type category are selected. Multiple selection is possible.

Order group

Only operations of orders of a specific order group are selected. Multiple selection is possible.

MRP controller

Only operations of orders of a specific MRP controller are selected. You can also use wildcards.

Customer name/designation

Only operations of orders for a specific customer are selected. You can also use wildcards.

Sales order

Only operations of the order that matches the specified sales order are selected. You can also use wildcards.

Project number

Only operations of the order with the specified project number are selected. You can also use wildcards.

Planned order

Only operations of the order that matches the specified planned order are selected. You can also use wildcards.

Operation status

The system only selects operations of a specific operation status. Multiple selection is possible.

Control

The system only selects operations that have a status with a specific control indicator. Multiple selection is possible.

Check responsibility area

Using this option, the user can specify if the system checks the responsibility area of the workplace or the responsibility area of the object operation/order to display data. To use this selection option, you require the function authorization chkresp.



This selection option is only available, if you enable the extension [scec2](#).

Data collection

Depending on the option selected, the check is performed against the planned dates/the scheduled dates/the basic dates (specification as date) of the current dataset.

The values that are calculated are specified by the

- selection of the tab *Planned dates*, *Scheduled dates* or *Basic dates*.

The values *Planned start* or *Planned end* described in the tables below are used as some kind of placeholders for the different combinations (current planned dates/planned dates, current planned dates/scheduled dates, current planned dates/basic dates).



To calculate the days, the system uses the specified dates and synchronizes them with the Gregorian calendar. The displayed format is: days.hours:minutes:seconds.

Deviation from start date

Earliness

Condition 1: Control	Condition 2: Date	Calculation *)
L, U, E, A	OP actual start < planned start	ABS (OP actual start minus planned start)

Lateness

Condition 1: Control	Condition 2: Date	Calculation *)
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Condition 1: Control	Condition 2: Date	Calculation *)
L, U, E, A, F	OP actual start > planned start	ABS (OP actual start minus planned start)
S, V	OP planned start > today	ABS (OP planned start minus today)

On-time delivery

On-time delivery = earliness + lateness

Deviation from end date

Earliness

Condition 1: Control	Condition 2: Date	Calculation *)
E, A	OP actual end < planned end	ABS (OP actual end minus planned end)

Lateness

Condition 1: Control	Condition 2: Date	Calculation *)
S, V, L, U, F	Today > planned end	ABS (today minus planned end)
E, A	OP actual end > planned end	ABS (OP actual end minus planned end)

On-time delivery

On-time delivery = earliness + lateness

*) The time is always specified as absolute value (ABS).

Field descriptions

When the data is requested, the table shows the order/operation, OP designation and article and additionally the basic dates, the dates of detailed planning and the actual dates.

Deviations from scheduling (earliness, lateness, on-time delivery) are displayed in format days.hours:minutes:seconds.



To calculate the days, the system uses the specified dates and synchronizes them with the Gregorian calendar.

Detail application: Schedule controlling (graphic)

The detail application Schedule controlling (graphic) shows the earliness, lateness and on-time delivery for all operations selected in the table. If no operation is selected, it will be interpreted as "all operations selected".

Display options

Display	Relating to	Group by	Consideration
☒ Delays	☐ Start	☒ Workplace	☒ Mean value/standard deviation
☐ Earliness	☒ End	☐ Group	☐ Total of durations
☐ On-time delivery		☐ Cost center	

Display:

You can specify whether the graphic shows earliness, lateness or on-time delivery performance.

Relating to:

You can specify whether the graphic shows the deviations from the start (planned start/actual start) or from the end (planned end/actual end) of the operation.

Group by:

The selection made specifies if the graphic shows the totaled data for the workplace, group or cost center.

Consideration:

Specifies whether

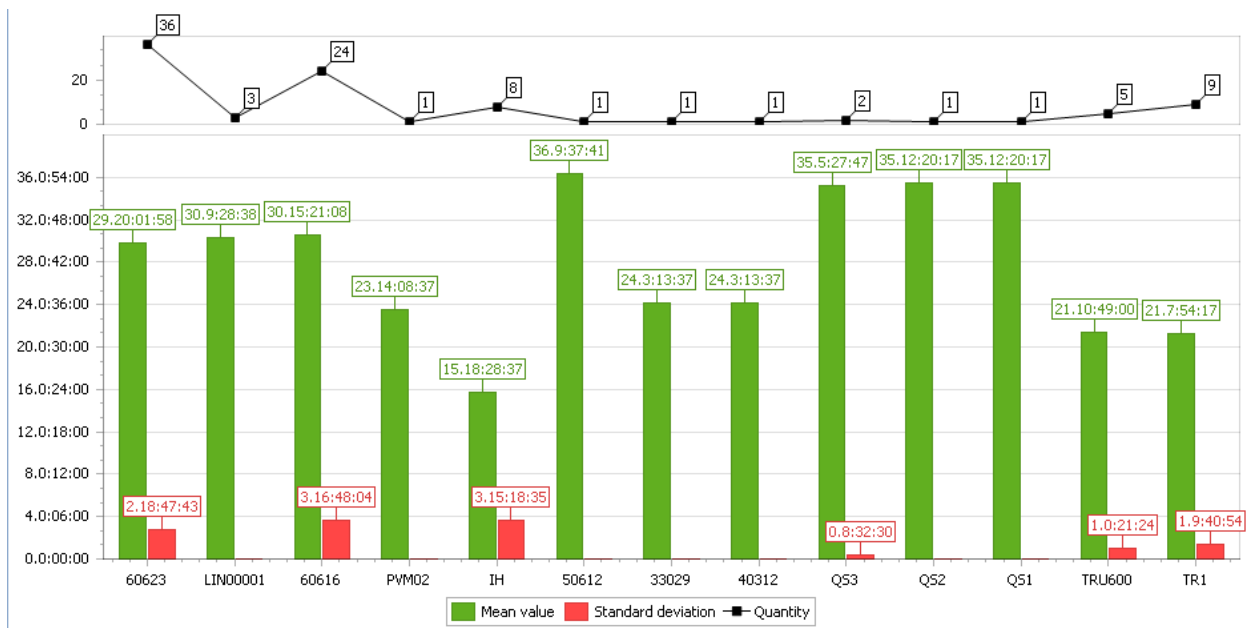
- the mean value AND the standard deviation

or

- the total of durations (lateness, ...)

are displayed.

Display



Data is displayed in format Days.Hours:Minutes:Seconds.

Irrespective of the selected display options, the number of operations used for the evaluation is always displayed in form of a line.

Calculations

Mean value

The mean value is calculated as follows:

$$\mu = \frac{\sum_{x=1}^n Earliness_x}{n}$$

$$\mu = \frac{\sum_{x=1}^n Delay_x}{n}$$

$$\mu = \frac{\sum_{x=1}^n Earliness_x + Delay_x}{n}$$

n = Number of OPs produced too early

Standard deviation

Standard deviation is calculated as follows: $\sigma = \sqrt{Var}$

In case of lateness, the variance is calculated as follows: $Var = \frac{\sum_{x=1}^n (Delay_x - \mu)^2}{n}$ (analog calculation for earliness or on-time delivery).

Toolbar

The parameters to call the function or target application are generally transferred from the table. For this reason, you should always select an entry before calling an application.



Order information (function authorization: orin)

Use this button to call the application [Order information](#).



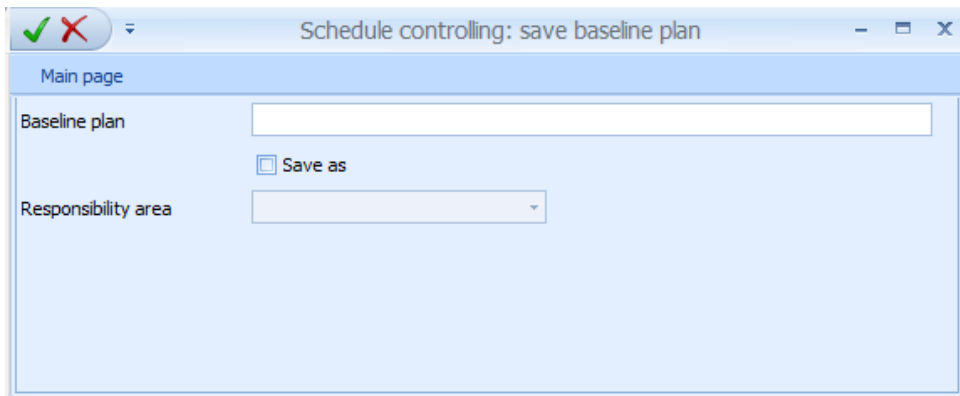
Order overview (function authorization: orov)

Use this button to call the application [Order overview](#).

Saving baseline plans

Function authorization: esvb; license: BDE-CAB

The applications *Operations* and *Pool of orders* provide this function. You can use this function to save baseline plans for selected operations that are used as selection criterion to identify deviations from planned dates.



The screenshot shows a Windows-style dialog box titled "Schedule controlling: save baseline plan". The dialog has a standard title bar with a green checkmark and a red X icon on the left, and minimize, maximize, and close buttons on the right. Below the title bar is a tab labeled "Main page". The main content area contains two input fields: "Baseline plan" and "Responsibility area". The "Baseline plan" field is a text box. The "Responsibility area" field is a dropdown menu. Between these two fields is a checkbox labeled "Save as".